SIMATS ENGINEERING

ASSESSMENT TOOL 2 (APPLICATION BASED PROBLEM SOLVING)

COURSE CODE:CSA09 **COURSE NAME: Programming in JAVA**

COURSE OUTCOME COVERED

CO5. *Construct database-driven web applications by implementing JDBC connectivity to perform SQL operations such as insert, update, and delete using different statement types.* (BL3, BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 1 **Total Marks:50**

Annexure A

APPLICATION BASED PROBLEM SOLVING

Code of Conduct:

I Harini G/192524011 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Harinigopi

Annexure A

APPLICATION BASED PROBLEM SOLVING

1: Smart Library Book Registration System (Statement + PreparedStatement)

A college library is digitizing its book inventory. The librarian needs a Java application that allows new books to be added while preventing duplicate ISBN numbers from being stored.

Database Table: Books

BookID Title Author ISBN AvailableCopies

Tasks:

1. Establish JDBC connectivity with MySQL.
2. Insert five book records using PreparedStatement.
3. Before insertion, verify whether the ISBN already exists using a Statement.
4. Display an appropriate success or duplicate message.

Problem-solving focus: Choosing between Statement and PreparedStatement for validation and insertion.

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Analysis & Solution Design	20%	Accurately analyzes the application scenario and designs a complete JDBC-based solution with appropriate SQL operations.	Correctly analyzes the scenario with minor gaps in the solution design.	Partially understands the scenario and proposes an incomplete solution.	Unable to identify the problem requirements or design a suitable solution.
JDBC Connectivity & Statement Usage	25%	Correctly establishes database connectivity and effectively uses the appropriate JDBC statement type (`Statement`, `PreparedStatement`, or `CallableStatement`).	Connection and statement implementation are mostly correct with minor errors.	JDBC connectivity is partially implemented; statement usage is inconsistent.	Fails to establish database connectivity or use JDBC statements correctly.
SQL Operations & Data Integrity	20%	Implements INSERT, UPDATE, and DELETE operations accurately while maintaining data integrity and proper parameter handling.	SQL operations are correct with minor logical or syntax issues.	Performs basic SQL operations with noticeable errors in data handling.	SQL operations are incorrect or fail to modify the database as required.
Program Logic & Exception Handling	20%	Demonstrates strong logical reasoning, validates inputs, and handles exceptions effectively using appropriate techniques.	Good program logic with basic input validation and exception handling.	Limited logical flow with minimal validation or error handling.	Poor program logic with no meaningful exception handling.
Code Quality & Output Verification	15%	Produces clean, modular, well-documented code and correctly verifies database results through meaningful output.	Code is organized with minor documentation or output verification gaps.	Code structure is inconsistent; output verification is incomplete.	Code is poorly structured with incorrect or missing output verification.

QUESTION 1:

```
import java.sql.*;


public class Library {


   public static void main(String[] args) {


      try {
         // Connect to MySQL database
         Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/library",
            "root",
            "your_password"
         );


         System.out.println("MySQL connection successful!");


         // Statement for checking duplicate ISBN
         Statement st = con.createStatement();


         // PreparedStatement for inserting book records
         PreparedStatement ps = con.prepareStatement(
            "INSERT INTO books VALUES (?, ?, ?, ?, ?)"
         );
```

```
        // Insert five books
        checkBook(st, ps, 1, "Java Basics", "James", "ISBN101", 5);

        checkBook(st, ps, 2, "C Programming", "Dennis", "ISBN102", 4);

        checkBook(st, ps, 3, "Data Structures", "Mark", "ISBN103", 6);

        checkBook(st, ps, 4, "Computer Networks", "Andrew", "ISBN104", 3);

        checkBook(st, ps, 5, "Database Systems", "Korth", "ISBN105", 7);

        // Close connection
        con.close();

    } catch (Exception e) {
        System.out.println(e);
    }
}

// Method to check ISBN and insert book
static void checkBook(Statement st, PreparedStatement ps,
                      int id, String title, String author,
                      String isbn, int copies) throws Exception {

    // Check whether ISBN already exists
```

```
    String query = "SELECT ISBN FROM books WHERE ISBN = '" + isbn + "'";

    ResultSet rs = st.executeQuery(query);

    if (rs.next()) {

        // ISBN already exists
        System.out.println("Duplicate ISBN: " + isbn);

    } else {

        // Insert book using PreparedStatement
        ps.setInt(1, id);
        ps.setString(2, title);
        ps.setString(3, author);
        ps.setString(4, isbn);
        ps.setInt(5, copies);

        ps.executeUpdate();

        System.out.println(
            "Book inserted successfully: " + title
        );
    }
}
```

}

OUTPUT:

```
mysql> select * from books;
+--------+-------------------+--------+---------+-----------------+
| BookID | Title             | Author | ISBN    | AvailableCopies |
+--------+-------------------+--------+---------+-----------------+
|      1 | Java Basics       | James  | ISBN101 |               5 |
|      2 | C Programming     | Dennis | ISBN102 |               4 |
|      3 | Data Structures   | Mark   | ISBN103 |               6 |
|      4 | Computer Networks | Andrew | ISBN104 |               3 |
|      5 | Database Systems  | Korth  | ISBN105 |               7 |
+--------+-------------------+--------+---------+-----------------+
5 rows in set (0.00 sec)

mysql>
```